NATIONAL GBV CRISIS TECHNOLOGY • SOUTH AFRICA

(Sankofa Safe)

A dual-persona discreet safety shield engineered for South African GBV survivors. Disguised as PowerCalc SA (Municipal Prepaid Electricity & STS Token Calculator) or a store-themed launcher, masking a hardware-encrypted domestic abuse evidence vault and zero-latency emergency beacon.

PowerCalc SA Camouflage

Runs as an authentic prepaid electricity calculator (Ekurhuleni/Boksburg, City Power, Eskom). 0ms panic exit hides all traces. Zero recents card.

Life-Safety Beacon

One-touch 72-hour SOS dispatch via SMS/WhatsApp with battery & GPS coordinates telemetry even on R0.00 airtime balance.

Court-Admissible J88 Vault

AES-256-GCM encrypted incident diary with ECTA SHA-256 hashes, compliant with SA Domestic Violence Act Form 2.

HUMAN-CENTRED SAFETY UX • ADVERSARY EVASION

End-to-end user story illustrating covert incident logging and critical moment escalation under intimate partner surveillance:

PHASE 1 **Everyday Use**	**PowerCalc SA Facade (Prepaid Electricity Utility):** Survivor uses her phone normally. The app presents as 'PowerCalc SA' with an electric power bolt icon, computing kWh units and STS tokens for Ekurhuleni/Eskom. If inspected by an abusive partner, it appears to be an indispensable household utility.
PHASE 2 **Secret Entry**	**Biometric Evidence Diary & J88 Capture:** In private, survivor enters secret calculation (1234=, 8426=, or 0000=) or long-presses the STS Token button. BiometricPrompt authenticates fingerprint. She logs verbal threats, photos of bruising, and SAPS OB numbers—encrypted on-device with AES-256-GCM.
PHASE 3 **Threat Alert**	**Immediate Threat Escalation (Armed SOS):** A violent altercation begins. Survivor triggers the Quick-Access SOS. A 3-second tactile vibration countdown engages. If not cancelled, high-priority emergency SMS alerts with 72-hour live GPS tracks and battery telemetry dispatch to 5 guardians.
PHASE 4 **0ms Camouflage**	**Zero-Trace Instant Panic Camouflage:** Footsteps approach. Survivor hits the hardware Back button or Escape key. Safe Haven closes in **0 milliseconds**, scrubbing decrypted data from RAM and reverting to PowerCalc SA without task thumbnails.

ENGINEERING & ARCHITECTURE SPECIFICATIONS

Modularized Jetpack Compose & React 19 architecture using Manual DI Container and Injectable Services:

UI Layer (Compose & React 19)	MainActivity • MunicipalPrepaidCalculator • LauncherFacadeScreen • SafeHavenScreen Declarative UI driven by StateFlow and React. PowerCalc SA utility facade, store-themed launcher, and live UAT banner. Zero XML.
Domain & ViewModel (MVVM)	SafetyTriggerViewModel • SankofaViewModel • UatFeedbackManager Finite state machine managing SOS countdown, cancellation grace period, duress PIN 9999= redirection, and UAT feedback sync.
Dependency Injection (AppContainer)	AppContainer (Singleton) • SankofaApplication.container Lightweight, manual dependency injection container providing single instances of DAOs, hardware managers, and telemetry helpers with zero runtime reflection.
Data, Security & Cloud Sync	VaultEncryptionHelper (AES-256-GCM) • Room SQLite / IndexedDB • Vercel & Supabase AndroidKeyStore StrongBox/TEE keys, ECTA SHA-256 hashes, SMS dispatch engine, and Supabase RLS anonymous UAT feedback persistence.

LIVE DEMONSTRATION, UAT TESTING & EVALUATION ACCESS

Pre-configured credentials to evaluate PowerCalc SA camouflage unlock, biometric vault, and emergency lifelines:

PowerCalc SA Secret Triggers

- **Calculator PINs:** 1234=, 8426=, or 0000=
- **Secret Gesture:** Long-press on STS Token button
- **Duress Coercion Alarm:** PIN 9999= (fake voucher + silent SOS)
- **Panic Reset:** Hardware Back key, Escape, or Quick-Exit

UAT Field Test & Emergency Lifelines

- **UAT Simulated Guardians:** 082 000 0001 / 082 000 0002
- **GBV Command Centre:** 0800 428 428 (Toll-Free, 24/7)
- **SAPS Police Flying Squad:** 10111
- **Zero-Airtime USSD Callback:** *120*7867# / *134*7355#

Vercel Cloud Deployment & Direct Mobile Downloads

- **Web PWA Hosted Access:** Desktop PC/Mac & iOS Safari zero-trace portal on Vercel Edge
- **Direct Mobile APK:** web/public/downloads/Umkhuseli_Safe_latest.apk (direct clean download)
- **UAT Testing Mode:** Built-in toggle intercepts live SOS to simulate police dispatch without escalating

LEAN STARTUP METHODOLOGY • PUBLIC BENEFIT MODEL

Social enterprise strategy addressing South Africa's femicide crisis via sustainable, zero-trace technology:

Problem • SA femicide 5x global avg. • Abusers inspect phones; safety apps cause retaliatory attacks. • Handwritten evidence dropped by court. • Data poverty prevents cloud apps.	**Solution** • PowerCalc SA prepaid utility facade. • Offline AES-256-GCM diary & ECTA hashes. • One-touch 72-hr GPS beacon. • Auto-generates Form 2 Court Affidavits. • Direct clean-domain APK downloads.	**Unique Value Proposition** *'Safety without suspicion; evidence without leakage.'* Dual-persona covert protection engineered for South African legal & socio-economic realities.
Unfair Advantage • Zero-airtime USSD (*120*7867#). • Proprietary 0ms camouflage exit. • Domestic Violence Act Form 2 & J88 medical exam prep integration. • Built-in UAT device testing suite.	**Channels & Distribution** • Thuthuzela Care Centre referrals. • University campus safety offices. • Police forums (CPFs) & NGOs. • Vercel clean web portal distribution.	**Key Metrics & Sustainability** • 0ms camouflage trigger rate (100%). • SOS delivery reliability (>99.5%). • 238/238 automated tests passing. • Public grants & Corporate CSI.

SECURITY SOFTWARE DEVELOPMENT LIFECYCLE • POPIA COMPLIANCE

Engineered according to OWASP Mobile Top 10 and South African POPIA & Domestic Violence Act mandates:

Cryptographic Protections

- **Key Generation:** 256-bit AES keys inside Android Hardware Keystore (StrongBox / TEE).
- **Cipher Mode:** Authenticated AES/GCM/NoPadding with unique 12-byte IVs per record.
- **At-Rest Storage:** Plaintext narratives and injuries never written unencrypted to disk.
- **Tamper-Evidence:** ECTA Section 15 SHA-256 cryptographic hashing.

Stealth & Memory Hardening

- **Zero-Latency Camouflage:** ViewModel state reset clears decrypted strings in 0 ms.
- **ProGuard / R8 Obfuscation:** Code and symbol obfuscation against decompilation.
- **Zero Third-Party Trackers:** 100% free of advertising SDKs and location telemetry.
- **Section 4(2) Address Withholding:** Complainants can withhold residential address.

Court Admissibility Guarantee (Act 116 of 1998 & Act 14 of 2021 Form 2 Compliance)

Under the Domestic Violence Amendment Act 14 of 2021 and ECTA Section 15, digital evidence is fully recognized by South African courts. Umkhuseli auto-compiles timestamped incident numbers, perpetrator names, impact summaries, J88 medical notes, and SAPS Occurrence Book references into standardized Form 2 affidavits for immediate submission to the Clerk of the Court.

DEPLOYMENT DELIVERABLES • VERIFIED RELEASE ARTIFACTS

Production-ready Gradle build configuration, verified release artifacts, and deployment pathways:

Compilation & Build Pipeline

• **Gradle Build Commands:**
./gradlew assembleDebug
./gradlew assembleRelease
./gradlew testDebugUnitTest (79/79 unit & Robolectric tests passing)
• **Compiled APK Artifact:**
web/public/downloads/Umkhuseli_Safe_latest.apk (25.1 MB signed)
• **GitHub Repository:**
github.com/OhhttooeXCIV/Umkhuseli_The_Protector

Quality Scorecard & Distribution Matrix

• **Target Platform:** Android 7.0 (API 24) to Android 16 (API 36)
• **Web Decoy Platform:** React 19 + TypeScript (159/159 Vitest tests passing)
• **Total Test Coverage:** 238/238 automated tests passing across 25 suites
• **Cloud Deployment:** Vercel Edge configuration with SPA routing
• **Direct Mobile Downloads:** Hosted web APK delivery for clean installs
• **Live UAT Ready:** In-app field testing mode with simulated dispatch intercept